Internship report on

Synthesis and Characterization of $NaLa(MoO_4)_2:Er^{3+},Yb^{3+}$ Nanomaterials and their Upconversion photoluminescence Properties



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Abstract

Nanomaterials of $NaLa_{1-x}(MoO_4)_2:x\%Er^{3+};10\%Yb^{3+}$(x=1,3,5,7,10) and $NaLa_{1-y}(MoO_4)_2:3\%Er^{3+};y\%Yb^{3+}$ (y=1,2,3,6,8,10) have been synthesized successfully through co-precipitation method at room temperature. The synthesized phosphors were crystalline and showing phase purity. Their photoluminescence up-conversion properties have been investigated under the excitation of 980 nm laser radiation. The up-conversion emission bands have been observed in the green region at 525 and 552 nm. And the other emission band appeared at 652 nm in the red region. Optimum up-conversion luminescence was obtained in $NaLa(MoO_4)_2$ nanomaterials doped with 3% Er^{3+} and 8% Yb^{3+}.

1. Introduction:

In recent time photoluminescence and energy transfer mechanism of rare earth doped double molybdate and tungstate have been a keen interest due to their enormous application in wide range including optoelectronic devices, bioimaging, biolabelling, biodetection assays, amplifier for fiber optics communication, solid state laser, visual display, finger print detection, cathode ray and fluorescent lamps. The luminescence properties of the rare earth doped nanoparticles depend strongly on the dopant rare earth ions and the host lattice. The host lattice may significantly influence the radiative and non-radiative (multiphonon relaxation as well as energy transfer) properties, lead to entirely different luminescence behavior [1].

In past few decades, many rare earth codoped double molybdate and tungstate have been synthesized with general formula $MRe(WO_4)_2$ or $MRe(MoO_4)_2$ (M=Alkali metalion and R=trivalent rare earth ion). Substitution of rare earth ions as luminescent centers in host matrices has removed limitation in traditional luminescent materials. Several efforts have been made to develop synthesis process for these rare earth doped double molybdate like solid state reaction, sol-gel method, co-precipitation method, hydrothermal method and microwave assisted hydrothermal method [2].

Usually double molybdate are prepared by solid state method that require high temperature, lengthy heating process and subsequent grinding which requires lot of labor and time, due to these we may lose emission intensity. Up Conversion (UC) photoluminescence particles, a new generation phosphor materials have a strong ability to convert near-infra red radiation with low energy into visible radiation with high energy. The sol–gel process provides some advantages over the conventional solid-state method, including good homogeneity, low calcination temperature, small particle size and narrow particle size distribution optimal for good luminescent characteristics. However, the sol–gel process has a disadvantage in that it takes a long time for gelation [3].

As compared with usual method co-precipitation method have advantages of very short time, low cost, reduces human labor and high purity of final crystalline samples. Here we have synthesized $NaLa_{1-x}(MoO_4)_2$codoped with Er^{3+} (x=1,3,5,7,10) and $NaLa_{1-y}(MoO_4)_2$codoped withYb^{3+} (y=1,2,3,6,8,10). Among rare-earth ions, the Er^{3+} ion is suitable for the infrared to visible light conversion through the UC process due to a proper

electronic energy level configuration. The co-doped Yb^{3+} ions can remarkably enhance the UC yield due to the efficient energy transfer from Yb^{3+} to Er^{3+}. The Yb^{3+} ion, as a sensitizer, can be excited by incident infrared light source, and then the energy is transferred to the Er^{3+} ion activator from which the visible radiation can be emitted. The Er^{3+} ion activator is the luminescence center of the UC particles, whereas the sensitizer enhances the UC luminescence [2].The synthesized $NaLa(MoO_4)_2:Er^{3+}/Yb^{3+}$ were characterized by X-Ray Diffraction(XRD). Spectroscopic Properties were examined by Photoluminescence (PL) emission and ATR spectrophotometer.

1.1 Experimental Section:

1.2 Materials:

All the chemicals were analytical grade and were directly used directly without further purification. The $Yb(NO_3)_3.xH_2O$ (99.9%pure) was used as Ytterbium source, $Er(NO_3)_3.5H_2O$ (99.99%pure) as erbium source, $La(NO_3)_3.6H_2O$ as lanthanum source, $Na(MoO_4).2H_2O$ as molybdate source.

1.3 Synthesis:

White powder samples of $NaLa_{1-x-y}(MoO_4)_2:x\%Er^{3+}$; $y\%Yb^{3+}$ were prepared by co-precipitation method in which Er^{3+} and Yb^{3+} are expected to occupy La^{3+} sites in host lattice. Initially we started with 2.0615mmol of $La(NO_3)_3.6H_2O$ and 0.00629mol of $Na(MoO_4).2H_2O$,then we dissolved the $La(NO_3)_3.6H_2O$ with $Er(NO_3)_3.5H_2O$ and $Yb(NO_3)_3.xH_2O$ in one series of reaction where we are varying erbium (percentage in mmol , x=1,3,5,7,10) keeping ytterbium constant 20ml of Ethylene glycol by magnetic stirring and heating, after complete mixing the solution is cooled to room temperature naturally.

On the other side we prepared a solution of $Na(MoO_4).2H_2O$ in 5 ml of dionized water at room temperature. On adding dionized water solution of $Na(MoO_4).2H_2O$ in first solution of ethylene glycol we get a white precipitate, which is kept on magnetic stirring for two hours at room temperature. After completion of reaction the white precipitate is washed twice with ethanol, water and acetone. The same above procedure was repeated for the variation of ytterbium in host matrix keeping erbium fixed.

1.4 Characterization:

The phase composition of synthesized nanoparticles has been determined using a Bruker D8 Advance X-ray diffractometer (Ni-filtered Cu-K_α radiation, λ=1.05406 Å, V=40 kV, I=25mA). The FTIR spectra were performed in the range of 200-4000 cm^{-1} with 4cm^{-1} resolution using Bruker Vertex 70 series spectrometer with ATR mode. The room temperature up-conversion luminescence spectra of nano-materials were measured using Fluorolog 3-221 with a 980 nm laser diode (2.5 W) as excitation source.

2. Results and Discussion:

3.1 XRDAnalysis:

The X-Ray Diffraction (XRD) of $Na_{0.92-x}La(MoO_4)_2$:x%Er; 8%Yb and$Na_{0.97-y}La(MoO_4)_2$:3%Er; y%Yb are in Fig 1. It can be seen from the figure that the peak positions of the X-ray diffraction peaks of all products are basically consistent, which are consistent with the JCPDS card PDF#00-024-1103.

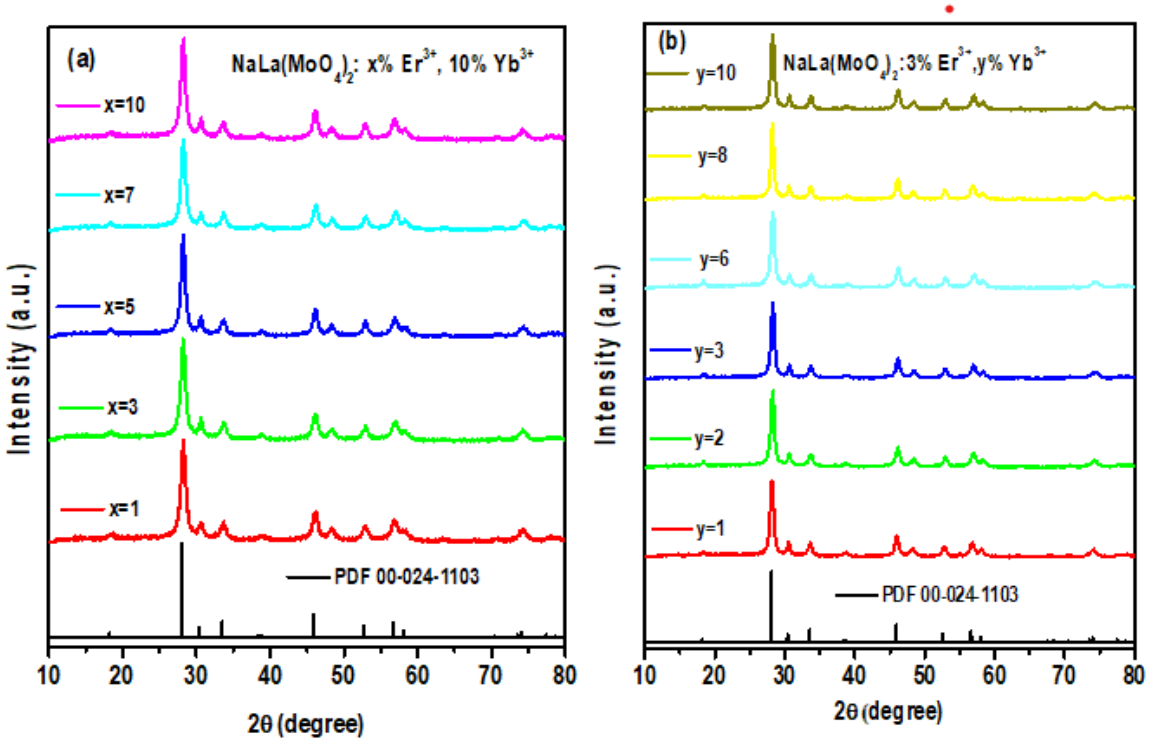


Fig1. XRD patterns of **(a)**$NaLa_{1-x}(MoO_4)_2$:10%Yb^{3+};x%Er^{3+} and **(b)**$NaLa_{1-y}(MoO_4)$2:3%Er^{3+};Yb^{3+}nanomaterials along with standard JCPDS card 00-024-1103.

On Comparing with the standard card of tetragonal phase of $NaLa(MoO_4)_2$,all diffraction peaks of the samples, can be well indexed to pure scheelite-type tetragonal phase of $NaLa(MoO_4)_2$with the space group of I41/a with unit cell parameters of a=b=5.34300 and c=11.74300. No peaks of any other phases or impurities were detected. The strong and sharp diffraction peaks indicate a good crystallinity of the prepared samples. And no additional peaks or other phases can be detected, which reveals a high phase purity of the as-obtained samples. After comparing with JCPDS card no 00-024-1103it can be said that Er^{3+} and Yb^{3+} are well substituted in tetragonal phase of $NaLa(MoO_4)_2$[4].

3.3 Fourier Transform Infra-red Spectroscopy:

From the fig.2 and 3, we can see that the FTIR spectrum of $NaLa(MoO_4)_2$ comprises vibrational bands at 404, 1633, 2338, 2360, with two broad bands with shoulders at 700, 750, 840, 914 and 3215, 3415 cm^{-1}. The vibrational band at 404 cm^{-1} corresponds to the asymmetric bending of MoO_4^{2-}tetrahedra. Vibrational bands at 700, 750 cm^{-1} and 840, 914 cm^{-1} are attributed to the symmetric and asymmetric stretching vibrations of $MoO4^{2-}$ tetrahedra, respectively. All the vibrational modes are very well matched with previous reports [11, 12].Vibrational bands at 1633 and 2338, 2360 cm^{-1} are attributed to O−H bending and symmetric, asymmetric stretching of C−H bonds of the ethylene glycol molecule present on the surface of the nanoparticles. The broad band at 3415 cm^{-1} with a shoulder peak at 3220 cm−1 corresponds to free and bounded O−H stretching frequencies of water molecules present on the surface of $NaLa(MoO_4)_2$ nanoparticles. These results shows that ethylene glycol molecules are attached to the surface of $NaLa(MoO_4)_2$nanoparticles [13].

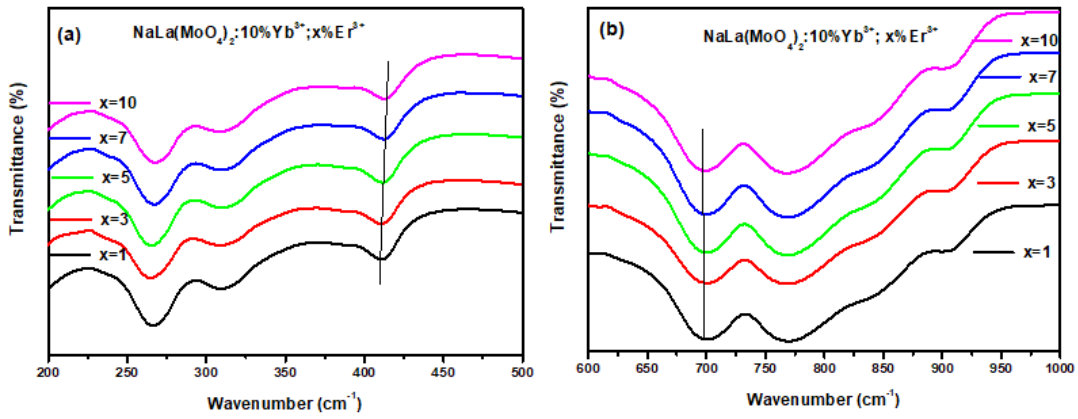


Fig2. ATR FTIR spectra of $NaLa(MoO_4)_2$:x%Er^{3+}, 8%Yb^{3+}nanomaterials in the **(a)** 200 - 500 cm^{-1} and **(b)**600 – 1000cm^{-1} regions.

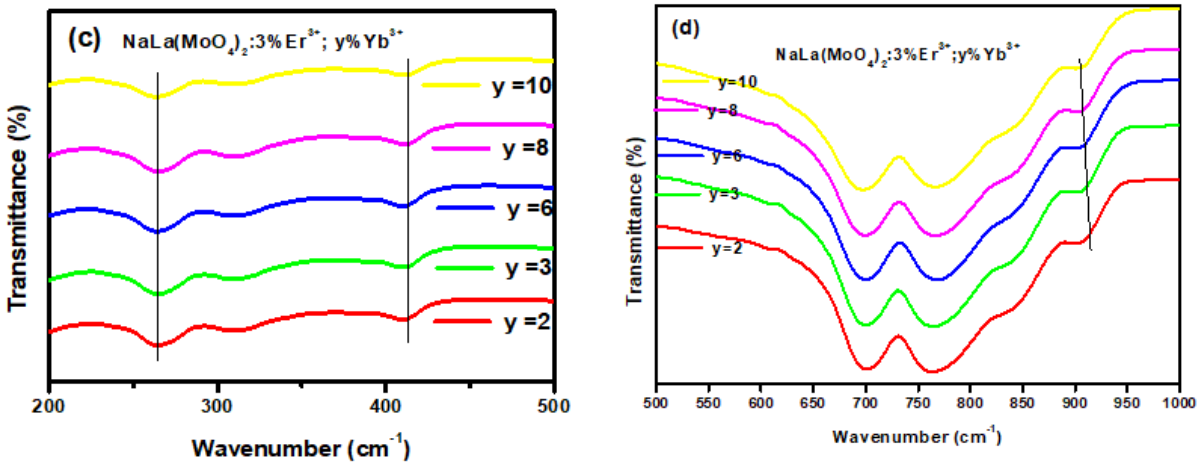


Fig3. ATR FTIR spectra of NaLa(MoO4)2:3%Er^{3+}, y%Yb^{3+} nanomaterials in the **(a)** 200 - 500 cm^{-1} and **(b)**600 – 1000 cm^{-1} regions.

3.3 Up Conversion Photoluminescence Properties :

Fig.4(a) and 5(a) shows the photoluminescence emission spectra of $NaLa_{1-x}(MoO_4)_2$:10%Yb;x%Er and$NaLa_{1-y}(MoO_4)_2$:3%Er;y%Yb respectively excited under 980nm at room temperature. The particles of both $NaLa_{1-x}(MoO_4)_2$:10%Yb;x%Er and $NaLa_{1-y}(MoO_4)_2$:3%Er;y%Yb exhibit a strong emission at 525nm and weak at 552nm in green region corresponding to the $^2H_{11/2}\rightarrow{}^4I_{15/2}$ and $^4S_{3/2}\rightarrow{}^4I_{15/2}$ transitions, respectively while a very weak 655nm emission band in red region corresponding to $^4F_{9/2}\rightarrow{}^4I_{15/2}$ transition.

While increasing the concentration of Yb^{3+} in $NaLa_{1-y}(MoO_4)_2$:3%Er;y%Yb the emission intensity increases and becomes maximum at 8% as shown in Fig.5(b), after that it decreases this effect is known as concentration quenching. Similarly if we vary the Erbium concentration the optimal concentration for Erbium is found to be 3% as shown in Fig.4(b). But when we are varying the Erbium concentration the emission intensity is much larger than the case when we are varying Ytterbium concentration this clearly indicates that there is possibility of energy transfer between Er^{3+} and Yb^{3+}, and the optimal concentration for the host matrix is 3%Er^{3+} and 8%Yb^{3+} for Strong green emission.

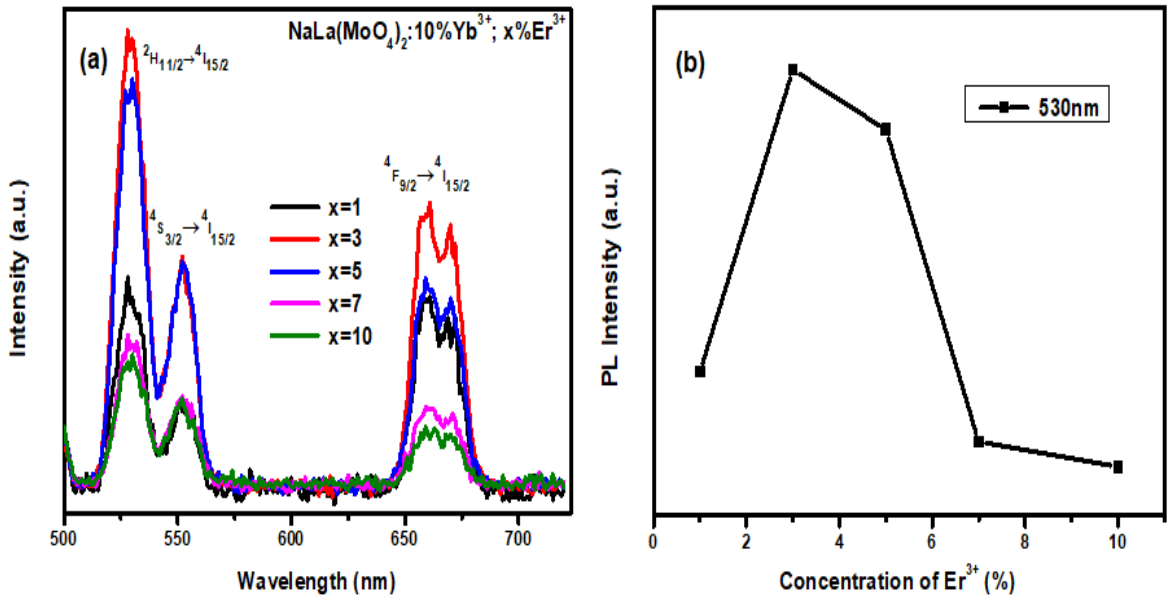


Fig.4 (a)PL emission spectra of $NaLa_{1-x}(MoO_4)_2$:10%Yb^{3+}, x%Er^{3+} (x=1,3,5,7,10) phosphors with different doping concentration excited at 980 nm, **(b)** Emission intensity of $^2H_{11/2} \rightarrow {}^4I_{15/2}$ transition as function of Er^{3+} concentration.

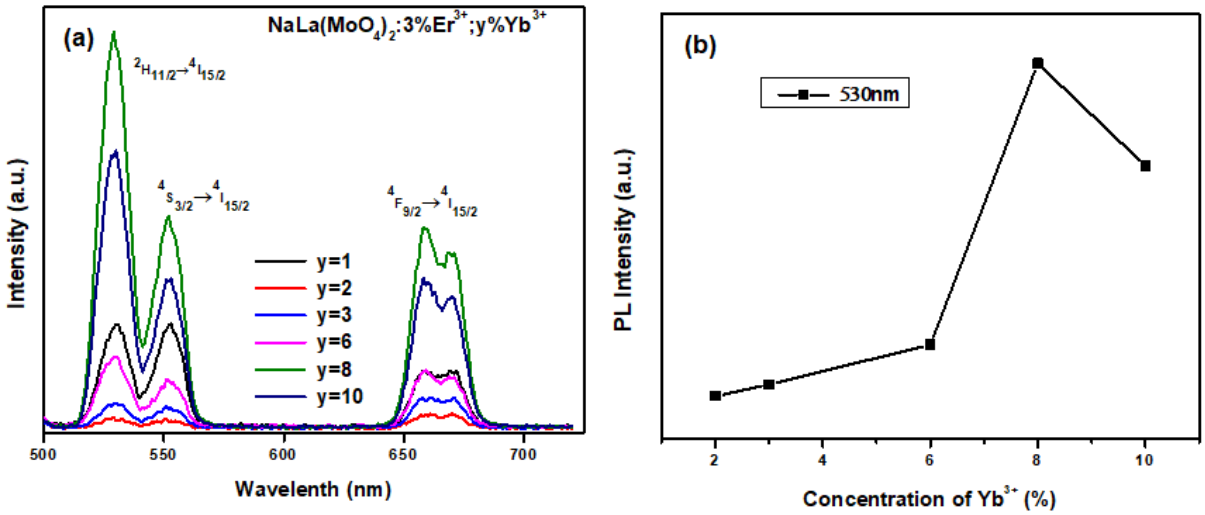


Fig.5(a) PL emission spectra of $NaLa_{1-y}(MoO_4)_2$:3%Er^{3+}, y%Yb^{3+} (y=1,2,3,6,8,10) phosphors with different doping concentration excited at 980-nm,**(b)** Emission intensity of $^2H_{11/2} \rightarrow {}^4I_{15/2}$ transition as function of Yb^{3+} concentration.

Now the energy transfer mechanism can be explained by the schematic energy level diagram of Er^{3+} ion (activator) and Yb^{3+} ion (sensitizer) in the as-prepared $NaLa(MoO_4)_2$:Er^{3+}/Yb^{3+} samples. For Er^{3+}/Yb^{3+} co-doped UC phosphors, the Yb^{3+} ion

sensitizer can be effectively excited by the energy of an incident light source, this energy is transferred to the activator, where radiation can be emitted.

The Er^{3+} ion activator is the luminescence center for these UC particles, and the sensitizer enhances the UC luminescence efficiency due to the energy gaps the $^2F_{7/2}$ and the $^2F_{5/2}$ of Yb^{3+}. Fig.6 (energy diagram) shows schematic energy level diagrams of Er^{3+} ions (activator) and Yb^{3+} ions (sensitizer) in the as-prepared $NaLa(MoO_4)_2:Er^{3+}/Yb^{3+}$ samples and the UC mechanisms accounting for the green and red emissions at 980 nm laser excitation. The UC emissions are generated through multiple processes of ground state absorption (GSA) and energy transfer (ET).

For the green emissions, under the excitation of 980 nm, the Yb^{3+} ion sensitizer is excited from the ground state of the $^2F_{7/2}$ to the excited state of the $^2F_{5/2}$ through GSA process, and transfers the energy to the excited Er3+ ions and promotes it from the $^4I_{15/2}$ to the $^4I_{11/2}$ by the ET process of $^4I_{15/2}(Er^{3+}) + {}^2F_{5/2}\ (Yb^{3+}) \rightarrow {}^4I_{11/2}\ (Er^{3+}) + {}^2F_{7/2}\ (Yb^{3+})$. Another Yb^{3+} ion at the $^2F_{5/2}$ level transfers the energy to the excited Er^{3+} ion, and then transits further the energy from the $^4I_{11/2}$ to the higher $^4F_{7/2}$ level by another ET process of $^4I_{11/2}\ (Er^{3+}) + {}^2F_{5/2}\ (Yb^{3+}) \rightarrow {}^4F_{7/2}(Er^{3+}) + {}^2F^{7/2}\ (Yb^{3+})$, which are for the population of a different level in Er^{3+}. The populated $^4F^{7/2}$ level relaxes rapidly and non-radiatively to the next lower $^2H_{11/2}$ and $^4S_{3/2}$ levels in Er^{3+} because of the short lifetime of the $^4F_{7/2}$ level.

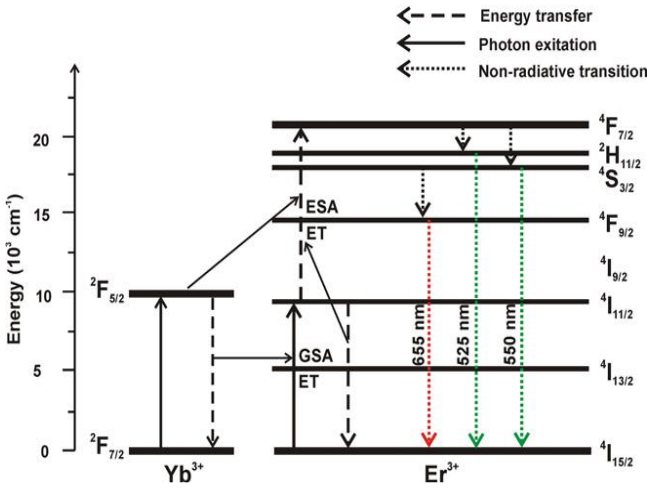


Fig6. Schematic energy level diagrams of Er^{3+} ions (activator)and Yb^{3+} ions (sensitizer) in the as-prepared $NaLa(MoO_4)_2:Er^{3+};Yb^{3+}$system and the up conversion mechanisms accounting for the green and red emissions under 980 nm laser excitation.

Then, the radiative transitions of the $^2H_{11/2} \rightarrow {}^4I_{15/2}$ and $^4S_{3/2} \rightarrow {}^4I_{15/2}$ processes can produce green emission at 525 and 550 nm. It is noted that the green up conversion luminescence can be induced by a two-photon process [5,6]. For red emission, the $^4F_{9/2}$ level is populated by non-radiative relaxation from the $^4S_{3/2}$ to the $^4F_{9/2}$ level and the second ET from the $^4I_{13/2}$ to the $^4F_{9/2}$ level in Er^{3+}. Finally, the $^4F_{9/2}$relaxesradiatively to the ground state at the $^4I_{15/2}$ level and releases red emissions at 655 nm [7, 8]. The strong 525 nm and 550 nm emission bands in the green region, as shown in Fig. 4are assigned to the $^2H_{11/2} \rightarrow {}^4I_{15/2}$ and $^4S_{3/2} \rightarrow {}^4I_{15/2}$ transitions of Er^{3+} ions, respectively, while the weak 655-nm emission band in the red region is assigned to the $^4F_{9/2} \rightarrow {}^4I_{15/2}$ transition.

The much higher intensity of the $^2H_{11/2} \rightarrow {}^4I_{15/2}$ transition in comparison with the $^4S_{3/2} \rightarrow {}^4I_{15/2}$ transition in Fig.6may be induced by the concentration quenching effect by the energy transfer between the nearest Er^{3+} and Yb^{3+} ions and the interactions between doping ions in the host matrix [9,10]. This means that the green band $^2H_{11/2} \rightarrow {}^4I_{15/2}$ transitions are assumed to be more easily quenched than the $^4S_{3/2} \rightarrow {}^4I_{15/2}$ transition by non-radiative relaxation in the case of the host matrix.

3. Conclusion:

$NaLa(MoO_4)_2:Er^{3+}/Yb^{3+}$up conversion phosphor with the Er^{3+} and Yb^{3+} doping were successfully synthesized by co-precipitation method. The UC photoluminescence and Energy transfer mechanism have been discussed in this report. Under the excitation at 980-nm, $NaLa_{1-y}(MoO_4)_2:10\%Yb^{3+};x\%Er^{3+}$ and $NaLa(MoO_4)_2:3\%Er^{3+}\%;y\%Yb^{3+}$ particles exhibited a strong 525 nm emission band, a weak 552nm emission band in the green region, which were assigned to the $^2H_{11/2} \rightarrow {}^2I_{15/2}$ and $^4S_{3/2} \rightarrow {}^4I_{15/2}$ transitions respectively, while a very weak 655nm emission band in the red region was assigned to the $^4F_{9/2} \rightarrow {}^4I_{15/2}$ transition. Optimum up-conversion luminescence was obtained in $NaLa(MoO_4)_2$ nanomaterials doped with $3\%Er^{3+}$ and $8\%Yb^{3+}$.

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